DS 6021: Introduction to Predictive Modeling

Lab 1: Statistical Learning and Experimental Decisions

Fall 2026 **30 Points**

Overview

A fast-food chain tested three marketing promotions for a new menu item. Your task is to determine what the experiment tells us about promotion effectiveness and how that information should guide a business decision.

The emphasis of this lab is on **statistical reasoning and interpretation**, not writing code from scratch.

Use of AI and Course Code

You **may use generative AI tools** and/or adapt code from class for the **coding components** of this lab.

You may use these tools to:

- generate Python syntax,
- adapt code demonstrated in class,
- debug code,
- create visualizations, and
- implement bootstrap or statistical procedures.

However, the **interpretations, explanations, decisions, and conceptual responses must reflect your own understanding**.

You are responsible for understanding any code you submit and for verifying that it answers the question being asked.

If you use generative AI, include one brief statement at the end of your notebook describing how you used it.

Data

The dataset contains weekly sales from stores assigned to one of three promotions.

Variable	Description
`MarketID`	Market identifier
`MarketSize`	Market size
`LocationID`	Store identifier
`AgeOfStore`	Store age in years
`Promotion`	Promotion: 1, 2, or 3
`week`	Week: 1–4
`SalesInThousands`	Weekly sales in thousands of dollars

Treat `Promotion` and `week` as categorical variables.

Learning Outcomes

By the end of this lab, you should be able to:

- distinguish inference from prediction;
- define and interpret a treatment effect;
- use resampling to quantify uncertainty;
- interpret confidence intervals in context;
- distinguish statistical evidence from practical importance;
- reason about experiments involving multiple treatments.

Part 1: Understand the Experiment [5 points]

Before performing any analysis, think about the study design.

1. **[2 pts]** Why is it important that this study was conducted as a **designed experiment**, with stores randomly assigned to promotions, rather than simply observing which promotions stores chose to use?
2. **[1 pt]** What is the experimental unit, treatment, and outcome in this experiment?
3. **[2 pts]** The company wants to determine whether promotion strategy **affects** sales.

   Is this primarily an **inference** or **prediction** problem? Explain why.

Part 2: Explore Before You Test [4 points]

Use appropriate summaries and visualizations to examine sales across the three promotions.

1. **[1 pt - coding]** Create one useful visualization of `SalesInThousands` across the three promotions.
2. **[2 pts]** Describe what the visualization suggests about:
   - differences in typical sales across promotions, and
   - variation within each promotion.
3. **[1 pt]** Based only on this exploratory analysis, can you conclude that one promotion **caused** higher sales? Why or why not?

Part 3: A/B Testing [8 points]

Suppose the company first wants to compare only **Promotion 1** and **Promotion 2**.

Let

$$\theta = \mu_2 - \mu_1,$$

where μ_1 and μ_2 are the population mean weekly sales under Promotions 1 and 2.

1. **[2 pts]** Explain what θ represents in the context of this experiment.

   What would each of the following mean?

$$\theta > 0, \qquad \theta = 0, \qquad \theta < 0.$$

2. **[1 pt - coding]** Calculate

$$\widehat{\theta} = \bar{x}_2 - \bar{x}_1.$$

3. **[1 pt]** Interpret $\widehat{\theta}$ in the units of the original business problem.

   Recall that sales are measured in **thousands of dollars**.

4. **[1 pt - coding]** Use bootstrap resampling with at least 5,000 repetitions to construct a 95% confidence interval for

$$\mu_2 - \mu_1.$$

5. **[2 pts]** Interpret the confidence interval in context. Your response should describe both the **direction** and **magnitude** of plausible effects.

6. **[1 pt]** Suppose your confidence interval contains 0. Explain why this does **not** necessarily mean that the two promotions have exactly the same effect.

Part 4: Statistical Evidence vs. Business Value [5 points]

Suppose Promotion 2 requires additional advertising costs.

The company decides that Promotion 2 would only be worth adopting if it increases average weekly sales by at least

$2,000

per store. That is, a difference in sales of $2 in `SalesInThousands`.

Using your estimate and confidence interval from Part 3, answer the following.

1. **[1 pt]** Is there evidence that Promotion 2 increases average sales relative to Promotion 1?

2. **[2 pts]** Is there convincing evidence that the increase is at least $2,000 per week?

   Explain how your confidence interval helps answer this question.

3. **[2 pts]** Make one recommendation:

   Adopt Promotion 2 **Keep Promotion 1** **Collect More Data**

   Defend your recommendation using both the estimated effect and its uncertainty.

Part 5: Guided Exploration — Three Promotions [5 points]

The company actually tested three promotions.

The broader question is now:

Do average sales differ among Promotions 1, 2, and 3?

Guided Exploration.
We have not formally covered ANOVA and multiple comparisons yet (although slides and AB Testing codes from class includes this content).
You may use AI, course materials, or recommended textbook to help perform the computations.
Focus on understanding what the procedures tell us.

ANOVA

A one-way ANOVA considers

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

versus

$$H_A : \text{at least one population mean differs.}$$

Run a one-way ANOVA comparing sales across the three promotions.

1. **[2 pt]** The null hypothesis for the ANOVA is

$$H_0 : \mu_1 = \mu_2 = \mu_3.$$

   Explain in words what $\mu_1 = \mu_2 = \mu_3$ means in the context of this experiment.

2. **[2 pt]** If the ANOVA results do not support

$$\mu_1 = \mu_2 = \mu_3,$$

   what would be a natural next step to explore **which promotions** may differ? Be specific about the comparisons you would consider.

3. **[1 pt]** Why should we be more careful about false positives when making several comparisons rather than only one?

Part 6: From Inference to Prediction [3 points]

Suppose the company changes its question.

Instead of asking: Which promotion increases sales?

it asks: Can we predict next week's sales for a store?

1. **[1 pt]** Explain why this is now primarily a **prediction** problem.
2. **[1 pt]** Identify at least three variables in the dataset that might be useful predictors of sales.
3. **[1 pt]** How would you evaluate whether a predictive model is successful? How is that different from what we were trying to accomplish in the experiment?

Point Summary

Section	**Points**
Part 1: Understand the Experiment	5
Part 2: Explore Before You Test	4
Part 3: A/B Testing	8
Part 4: Statistical Evidence vs. Business Value	5
Part 5: Guided Exploration	5
Part 6: Inference vs. Prediction	3
Total	**30**

Of the 30 points, only **3 points** are awarded directly for producing code or computational output. The remaining points emphasize statistical reasoning, interpretation, and communication.

What to Submit

Submit one notebook containing your code, visualizations, numerical results, and responses. A link is preferred with enabled access.

If you used generative AI, include a brief statement at the end describing how you used it.